# Cerebro-Nojara D0 v1.24: code, materials, libraries, wireframes, tokens

PROPOSED code pack: full CSS, Three.js materials and lights, exact library versions, 9 canonical screens wireframed at phone 390 and desktop, and a token map. Not applied anywhere.

On this page


Status and sources
1. CSS and unified token sheet
2. Three.js renderer, materials, lights
3. Libraries
4. Wireframes
5. Screen-by-screen token map
GAP list


## Status and sources

⚠ **PROPOSED, delivered as a Page.**
The box disk is full and the shell timed out, so no PDF could be saved to CEREBRO_D0_v1.24_CODE_PACK_20261031. Nothing was pushed, merged, deployed or applied.

| | |
|---|---|
| CSS, tokens, materials, lights | Tarek paste t1192u, verbatim |
| Teal + AgX fallback | Grandy review notes |
| Libraries | CRB_NJR_26 package.json (version 1.28.0) |
| Screens S01–S09 | CRB_NJR_26 src/gui/visualStateMachine.ts |
| Camera numbers | CRB_NJR_26 src/gui/cameraPresets.ts |
| v1.1 rebuild + AI Studio zip | GAP: unreadable (disk full) |

# 1. CSS and unified token sheet

```
:root {
  --cj-bg-viewport: #1B2230;
  --cj-bg-surface: #101622;
  --cj-glass-panel: rgba(27, 34, 48, 0.88);
  --cj-gold-primary: #D4AF37;
  --cj-gold-sheen: #FFF4D0;
  --cj-silver-metal: #C5CCD6;
  --cj-slate-blue: #3A516E;
  --cj-border-hairline: rgba(197, 204, 214, 0.22);
  --cj-glow-accent: rgba(212, 175, 55, 0.35);
}
.cj-viewport-container {
  background: radial-gradient(circle at 50% 40%, #1B2230 0%, #101622 100%);
  color: #F2F6F8;
  overflow: hidden;
  touch-action: none;
}
.cj-glass-card {
  background: var(--cj-glass-panel);
  backdrop-filter: blur(14px);
  -webkit-backdrop-filter: blur(14px);
  border: 1px solid var(--cj-border-hairline);
  box-shadow: 0 16px 32px rgba(0,0,0,0.4), inset 0 1px 0
rgba(255,255,255,0.1);
}
/* Review note (Grandy): keep teal only for two states */
:root { --cj-state-teal: #78D3B2; } /* listening + Needs-your-OK only */
```

```
// src/tokens/colours.ts  (values exactly as pasted, t1192u)
export const D0_V124 = {
  viewportBackground: { base: '#1B2230', gradientEnd: '#101622',
ambientTint: '#0E141E' },
  metallicGold: { primary: '#D4AF37', specularHighlight: '#FFF4D0',
shadowTone: '#685218', sheen: '#E6CA65' },
  brushedSilver: { primary: '#C5CCD6', specularHighlight: '#FFFFFF',
shadowTone: '#404854' },
  deepSlateBlue: { base: '#3A516E', deep: '#1E2B3D', light: '#5A779C' },
  glassTransmission: { refractionTint: '#EAF2FF', edgeRim: '#A4CBFF' },
  stateTeal: '#78D3B2', // review note: listening + Needs-your-OK only
  bodyText: '#F2F6F8',
} as const;
```

```
} as const;
```

| Old token | Role | v1.24 value | Source |
|---|---|---|---|
| #65ACFF | blue accent | #D4AF37 gold-primary | t1192u |
| #4C8DFF | blue accent | #D4AF37 gold-primary | t1192u |
| #F2B66D | amber accent | #D4AF37 gold-primary | t1192u |
| similar secondary blues | metal / secondary UI | #C5CCD6 silver-metal | t1192u |
| #B4DDFF | glass specular | kept as glass specularColor (spec keeps it) | t1192u |
| #78D3B2 | listening / Needs-your-OK | kept, those 2 states only | review note |
| body text | text | #F2F6F8 | t1192u |

## 2. Three.js renderer, materials, lights

Complete module: renderer with AgX and ACESFilmic fallback, stripe bump map, three shared materials, the four lights, and dispose helpers.

```
// src/components/CosmographCanvas.tsx  — D0 v1.24 visual layer (PROPOSED,
not applied)
import * as THREE from 'three'; // three ^0.186.1 (package.json) — AgX
exists since r160

export function setupRenderer(canvas: HTMLCanvasElement):
THREE.WebGLRenderer {
  const renderer = new THREE.WebGLRenderer({ canvas, antialias: true,
alpha: false });
```

```
  renderer.setPixelRatio(Math.min(window.devicePixelRatio, 2));
  renderer.setClearColor(0x1B2230, 1);
  renderer.outputColorSpace = THREE.SRGBColorSpace;
  // Review note: fall back if AgX is missing
  const agx = (THREE as unknown as { AgXToneMapping?: THREE.ToneMapping
}).AgXToneMapping;
  renderer.toneMapping = agx ?? THREE.ACESFilmicToneMapping;
  return renderer; // keep a single canvas
}

// Procedural sinusoidal stripe bump map (created once)
function makeStripeBump(size = 256, stripes = 32): THREE.CanvasTexture {
  const c = document.createElement('canvas');
  c.width = size; c.height = size;
  const ctx = c.getContext('2d')!;
  const img = ctx.createImageData(size, size);
  for (let y = 0; y < size; y++) {
    for (let x = 0; x < size; x++) {
      const v = 0.5 + 0.5 * Math.sin((x / size) * stripes * Math.PI * 2);
      const g = Math.round(v * 255);
      const i = (y * size + x) * 4;
      img.data[i] = g; img.data[i + 1] = g; img.data[i + 2] = g; img.data[i
+ 3] = 255;
    }
  }
  ctx.putImageData(img, 0, 0);
  const tex = new THREE.CanvasTexture(c);
  tex.wrapS = tex.wrapT = THREE.RepeatWrapping;
  tex.colorSpace = THREE.NoColorSpace; // bump data, not colour
  return tex;
}

export interface D0Materials {
  goldRibbed: THREE.MeshPhysicalMaterial;
  opticalGlass: THREE.MeshPhysicalMaterial;
  titanium: THREE.MeshPhysicalMaterial;
  stripeBump: THREE.CanvasTexture;
}

let shared: D0Materials | null = null;

export function getD0Materials(): D0Materials {
  if (shared) return shared; // create once, share everywhere
  const stripeBump = makeStripeBump();
  const goldRibbed = new THREE.MeshPhysicalMaterial({
    color: 0xD4AF37, metalness: 0.92, roughness: 0.22,
    clearcoat: 0.3, clearcoatRoughness: 0.1,
    sheen: 0.6, sheenRoughness: 0.3, sheenColor: new THREE.Color(0xFFF4D0),
    bumpMap: stripeBump, bumpScale: 0.05,
  });
  const opticalGlass = new THREE.MeshPhysicalMaterial({
```

```
  const opticalGlass = new THREE.MeshPhysicalMaterial({
    metalness: 0, roughness: 0.04, ior: 1.48, transmission: 0.98,
thickness: 1.2,
    specularIntensity: 1, specularColor: new THREE.Color(0xB4DDFF),
    transparent: true, opacity: 0.95,
  });
  const titanium = new THREE.MeshPhysicalMaterial({
    color: 0xC5CCD6, metalness: 0.85, roughness: 0.18,
    clearcoat: 0.5, clearcoatRoughness: 0.1,
  });
  shared = { goldRibbed, opticalGlass, titanium, stripeBump };
  return shared;
}

export function setupLights(scene: THREE.Scene): THREE.Light[] {
  // remove previous lights first
  scene.children.filter((o) => (o as THREE.Light).isLight).forEach((l) =>
scene.remove(l));
  const ambient = new THREE.AmbientLight(0x2a3648, 1.2);
  const key = new THREE.DirectionalLight(0xfff5e6, 2.4);
key.position.set(15, 25, 20);
  const fill = new THREE.DirectionalLight(0x8cb4ff, 1.2);
fill.position.set(-20, -10, -15);
  const rim = new THREE.DirectionalLight(0xffffff, 1.8);
rim.position.set(0, 30, -20);
  const lights = [ambient, key, fill, rim];
  lights.forEach((l) => scene.add(l));
  return lights;
}

// Swap old materials without per-frame allocation
export function applyMaterial(mesh: THREE.Mesh, next: THREE.Material): void
{
  const prev = mesh.material as THREE.Material | THREE.Material[];
  mesh.material = next;
  const all = Array.isArray(prev) ? prev : [prev];
  const keep = shared ? [shared.goldRibbed, shared.opticalGlass,
shared.titanium] : [];
  all.forEach((m) => { if (!keep.includes(m as THREE.MeshPhysicalMaterial))
m.dispose(); });
}

export function disposeD0(): void {
  if (!shared) return;
  shared.goldRibbed.dispose(); shared.opticalGlass.dispose();
  shared.titanium.dispose(); shared.stripeBump.dispose();
  shared = null;
}
```

## 3. Libraries

| Library | Version | Type |
| --- | --- | --- |
| three | ^0.186.1 | dep |
| @types/three | ^0.186.0 | dep |
| react / react-dom | ^19.0.1 | dep |
| motion | ^12.23.24 | dep |
| vite | ^8.3.0 | dep |
| @vitejs/plugin-react | ^6.1.1 | dep |
| @tailwindcss/vite / tailwindcss | ^4.3.3 | dep / dev |
| @google/genai | ^2.4.0 | dep |
| @mediapipe/tasks-vision | ^1.1.1 | dep |
| firebase | ^13.1.0 | dep |
| express | ^4.21.2 | dep |
| dotenv | ^17.2.3 | dep |
| jszip | ^3.10.2 | dep |
| lucide-react | ^0.546.0 | dep |
| typescript | ^7.0.2 | dev |
| tsx | ^4.21.0 | dev |
| esbuild | ^0.25.0 | dev |
| autoprefixer | ^10.4.21 | dev |
| @types/node | ^22.14.0 | dev |

| | | |
|---|---|---|
| @types/react / react-dom | ^19.3.0 | dev |
| @types/express | ^4.17.21 | dev |
| Cosmograph library | GAP (not in package.json; CosmographCanvas.tsx is custom three.js) | — |
| ElevenLabs SDK | GAP (not in package.json) | — |
| Fonts IBM Plex Sans / Jaapokki | GAP (no package) | — |

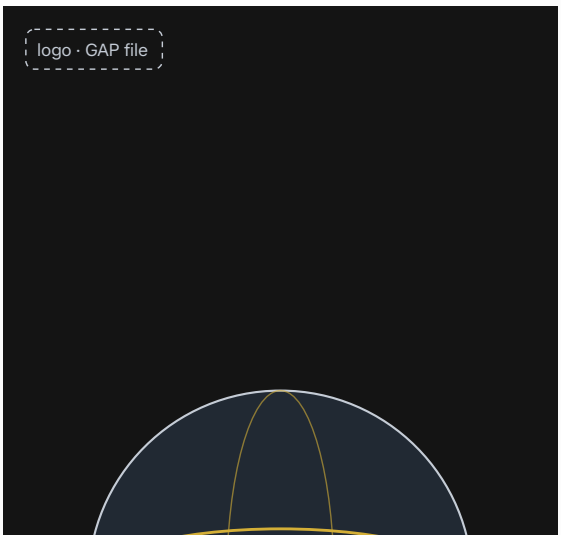
## 4. Wireframes

One screen per section, opening on S01 with the master sphere alone. Vector boxes in the v1.24 palette; layout follows only the repo description for each screen.

### S01 · Boot / Arrival

28mm overview, subtle sphere drift, voice pill idle, no inspector panels open. Camera universe: 28mm · 46.4° · 3200ms · 0.32 occupancy. State: idle.

Phone 390




Desktop 1280

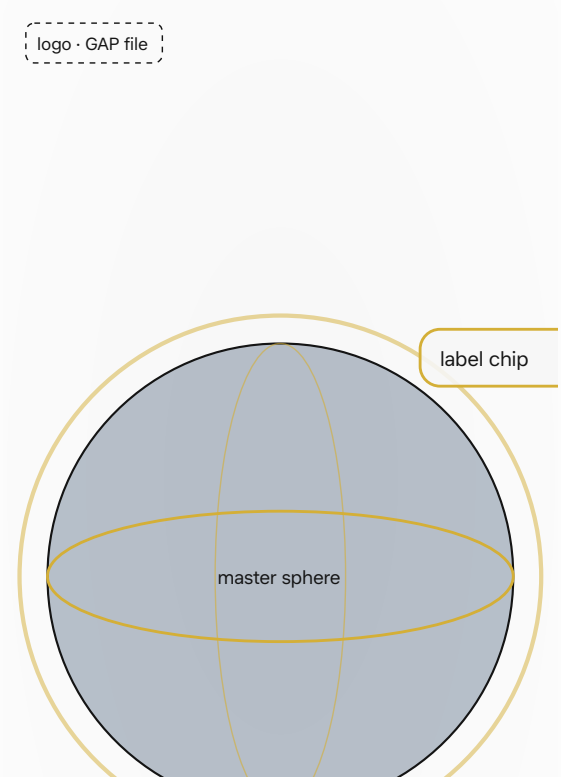




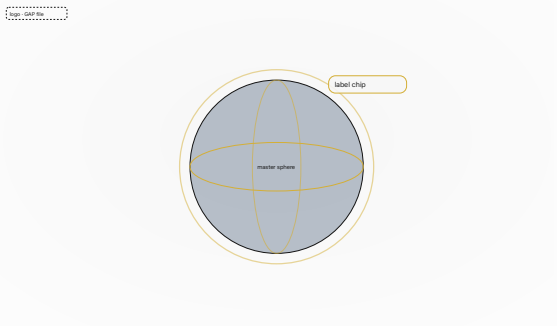

## S02 · Focus / Hover

Pointer halo and target ring highlight nearest station or sphere region; short label chip only. Camera approach: 35mm · 37.8° · 2400ms · 0.50 occupancy. State: focus.

### Phone 390



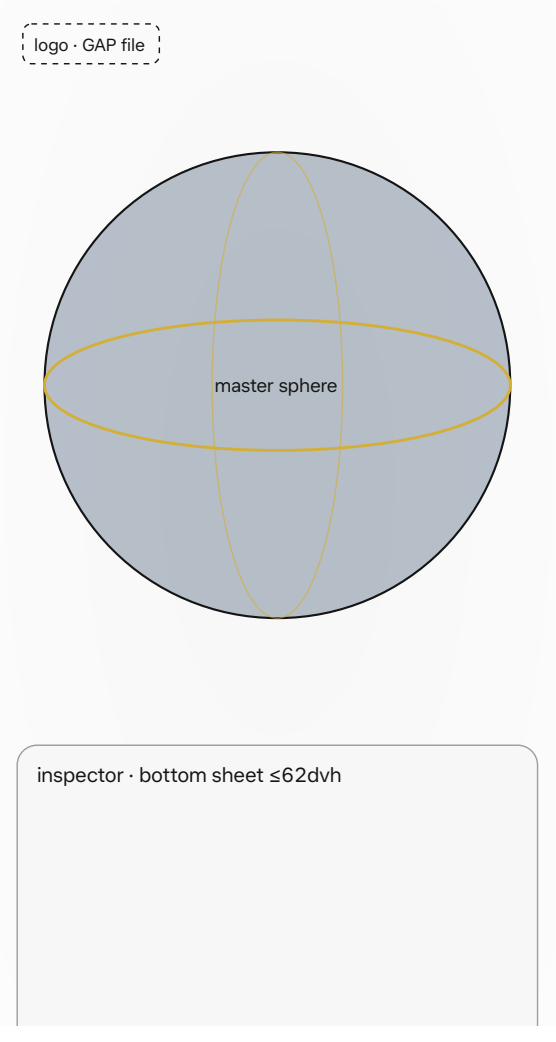

### Desktop 1280



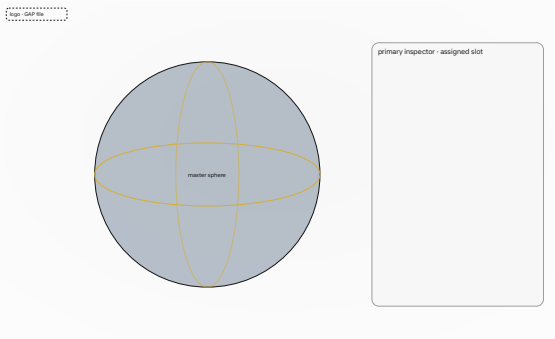
## S03 · Selected Target

Camera moves to 35-50mm framing; primary inspector opens in assigned slot without covering target. Camera focused: 50mm · 27.0° · 1800ms · 0.65 occupancy. State: selected.

### Phone 390



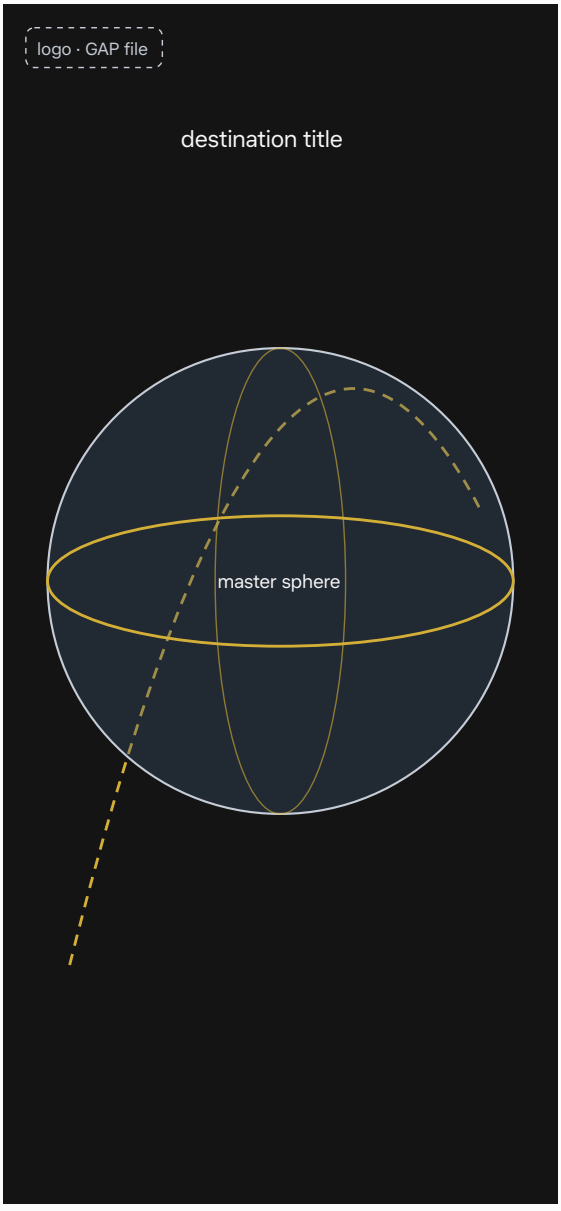

### Desktop 1280



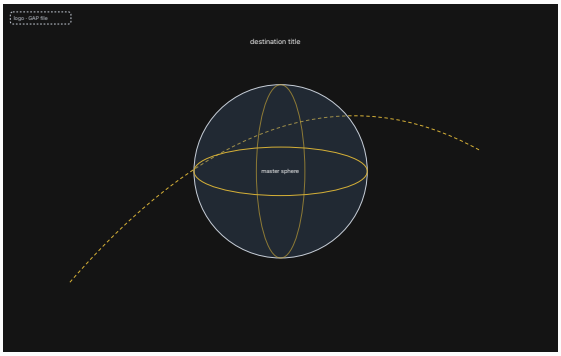
## S04 · Travel / Flight

Spline flight with route line, destination title, reduced panel opacity, no new popups during travel. Camera approach: 35mm · 37.8° · 2400ms · 0.50 occupancy. State: traveling.

Phone 390



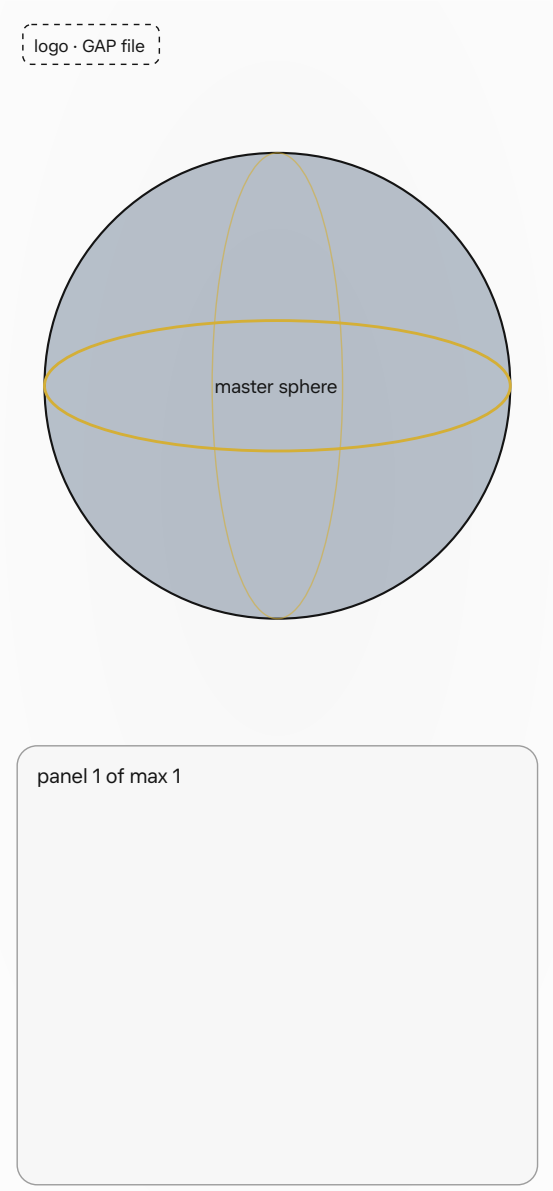

Desktop 1280



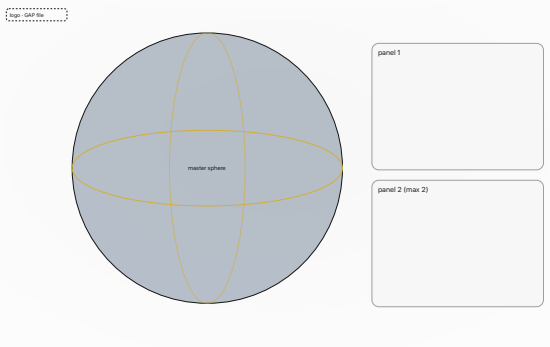
## S05 · Detail / Micro View

85mm close view of station cluster; max 2 panels on desktop, 1 on tablet/phone. Camera detail: 85mm · 16.1° · 1400ms · 0.78 occupancy. State: panel-open.

Phone 390




Desktop 1280

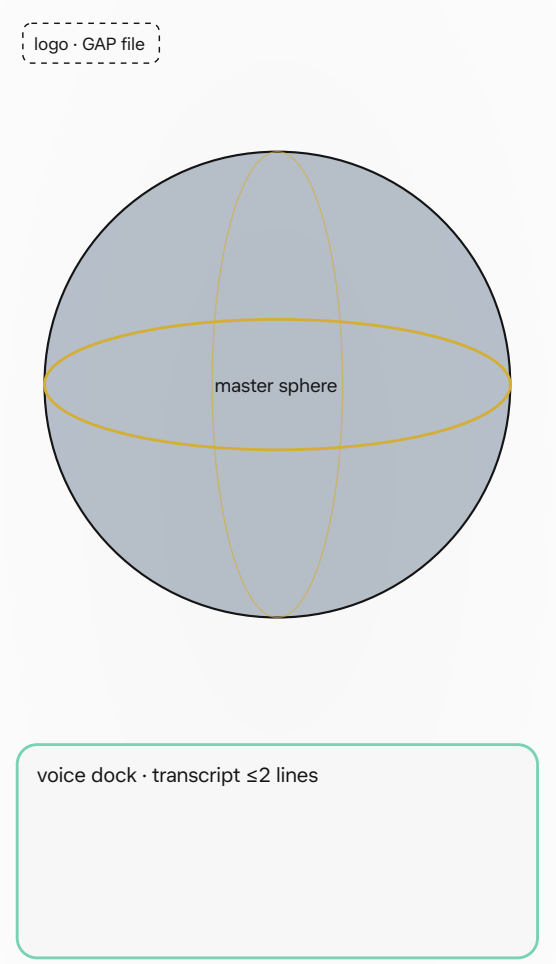



## S06 · Voice Active

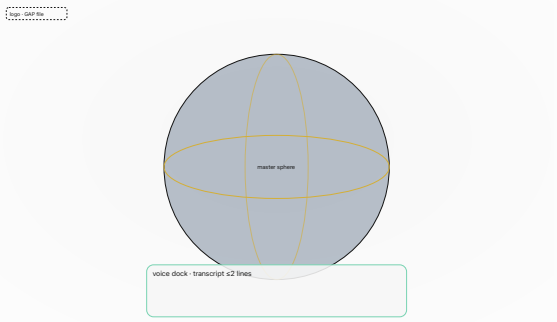
Voice dock expands transcript max 2 lines by default; full transcript opens as

drawer or tab. Camera focused: 50mm · 27.0° · 1800ms · 0.65 occupancy. State: speaking.

Phone 390




Desktop 1280

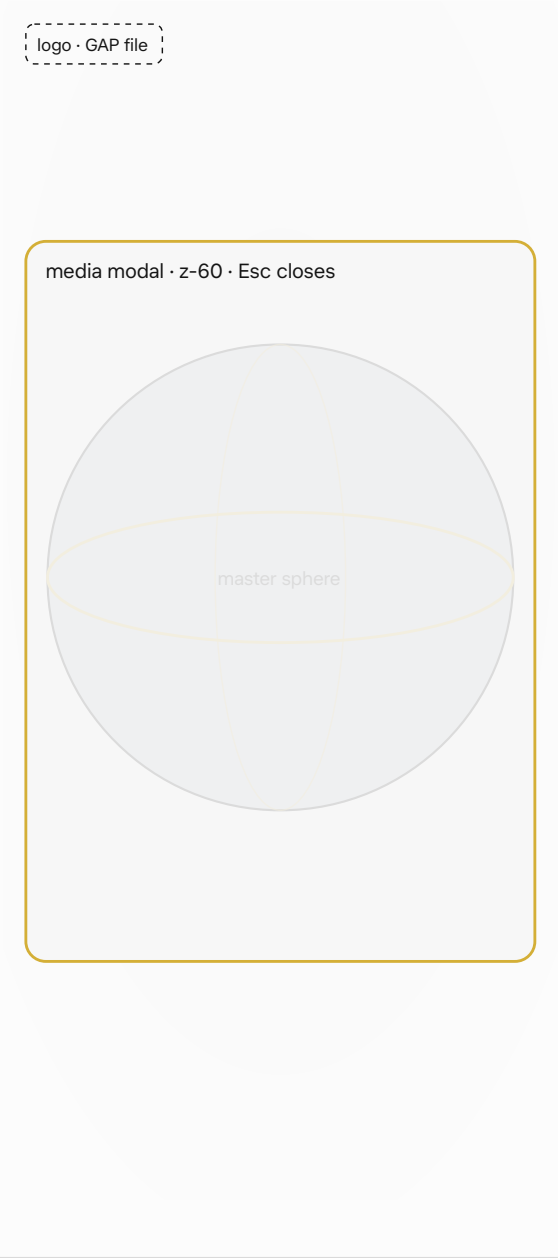


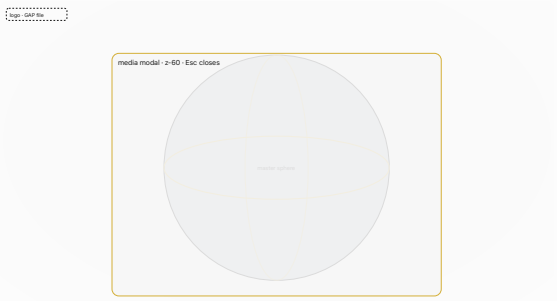

## S07 · Media / Presentation Modal

Centered modal at z-60; background panels dim and pause pointer capture; Escape closes. Camera focused: 50mm · 27.0° · 1800ms · 0.65 occupancy. State: panel-open.

Phone 390

Desktop 1280

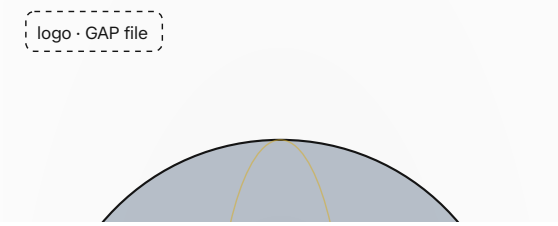


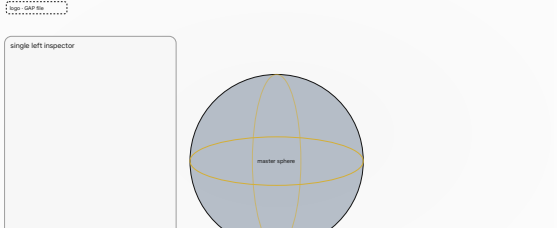

## S08 · Mobile / iPad Adaptive Mode

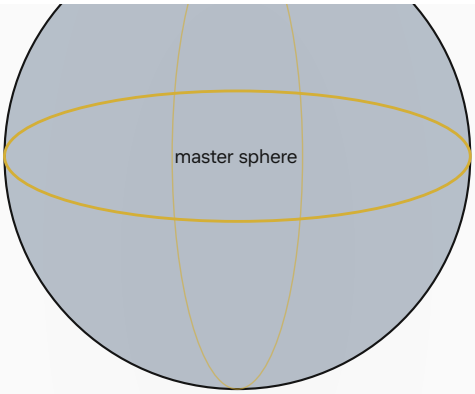
Single bottom sheet or single left inspector; touch orbit/pinch preserved on unobstructed canvas. Camera approach: 35mm · 37.8° · 2400ms · 0.50 occupancy. State: panel-open.

### Phone 390




### Desktop 1280






single bottom sheet

## S09 · Return / Resume

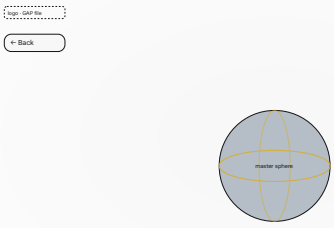
Back returns to previous camera bookmark and restores previous panel tab state. Camera universe: 28mm · 46.4° · 3200ms · 0.32 occupancy. State: landing.

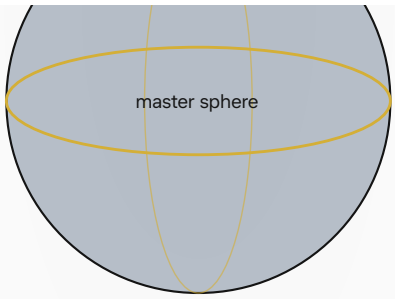
### Phone 390

logo · GAP file

← Back

### Desktop 1280






## 5. Screen-by-screen token map

| Screen | CSS tokens | Materials | Glass-card parts |
|---|---|---|---|
| S01 | viewport gradient, silver-metal | titanium sphere, gold ribbed rings, optical glass shell | voice pill (glass-card) |
| S02 | gold-primary, glow-accent | same + target ring (gold) | label chip (glass-card) |
| S03 | glass-panel, border-hairline | same | inspector (glass-card) |
| S04 | gold-primary route, panels reduced opacity | same | destination title |
| S05 | glass-panel ×2 / ×1 | same at 85mm | panels (glass-card) |
| S06 | teal #78D3B2 (listening only) | same | voice dock (glass-card) |
| S07 | glass-panel, dim | same, dimmed | modal (glass-card) |

| | | | |
|---|---|---|---|
| | scrim | | |
| S08 | glass-panel, border-hairline | same | bottom sheet / left inspector |
| S09 | silver-metal (Back) | same | Back control |

Mapping derived from the t1192u rules (glass-card on every panel, gold accents, silver secondary) applied to each repo screen description; it is a proposal, not read from code.

## GAP list

- v1.1 Gemini rebuild files (index.html, JSON): not read, box disk full
- AI Studio zip: not opened, shell down
- Logo / wordmark file for the wireframes
- Panel copy, labels and real media for every screen
- Exact panel pixel sizes beyond the repo rules (≤62dvh sheet is from v0.2 decisions)
- Cosmograph, ElevenLabs SDK and font package versions
- Rendered verification of the code (no build run)
- 2-minute idle film and particle-field specs: not in the sources used